Crisphine M. Ngari

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Research Interests

Analog circuit design · Mixed-signal VLSI design · Digital VLSI design · Programmable ASIC design · Semiconductor devices · Hardware and IoT security · Machine learning · LLM agents for hardware design · LLM agent security

Education

Southern Illinois University Carbondale Aug 2025 – Present
M.S., Electrical and Computer Engineering GPA: 3.89 / 4.0 *Carbondale, IL*

- Advisor: Prof. Ning Weng

GITAM School of Technology Sep 2020 – May 2024
B.Tech., Electronics and Communication Engineering First Class, GPA: 8.24 / 10 *Bangalore, India*

- Thesis: Wireless Localization on Constrained Hardware
- Study in India Merit-Based Scholarship (2020–2024)

Publications

Under Review

- N. C. Macharia, N. Weng. "Corroborating Telemetry: Resilient LLM-Driven Network Operations at the Cyber Tactical Edge." Under review, 2026.
- N. C. Macharia, N. Weng. "Towards Trustworthy IoT Network Security: Concept Bottlenecks and Differentiable Rules for Interpretable Open-Set Intrusion Detection." Under review, 2026.

Conference Papers

- E. Asogwa, N. C. Macharia, A. Bajpai, N. Telagam. "A Framework for Real-Time Localization in Constrained Devices Connected to the IoT." In *Proc. IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, 2024.

Research & Projects

Resilient LLM-Driven Network Operations at the Cyber Tactical Edge 2025 – Present
Southern Illinois University Carbondale

- Showed that a false diagnosis injected over unauthenticated syslog, SNMP, or gNMI steers an LLM network-operations agent into harmful reconfiguration on every trial, against 5% for the same payload written as a direct instruction.
- Established that taint-based telemetry sanitization cannot hold: masking device identifiers drops root-cause localization from 100% to 0%.
- Built *Ground-Truth Corroboration*, a deterministic verifier that admits a remediation-bearing claim only when live device state confirms it, cutting injection-attributable attack success to 0% at roughly 0.18 s per check across six models on an eight-router, multi-AS topology.

Interpretable IoT Intrusion Detection with Open-Set Robustness 2025 – Present
Southern Illinois University Carbondale

- Jointly evaluated Concept Bottleneck Models (CBMs) and Neuro-Symbolic NIDS (NeSy-NIDS) on IoT traffic classification using the CTU-IoT-23 and CIC-IoT-2023 datasets.
- Both architectures score open-set inputs by per-class Mahalanobis distance; neither incurs a classification accuracy cost.
- NeSy-NIDS reached an OOD AUROC of 0.906 on CTU-IoT-23 and a moderately regularized JointCBM 0.895, both at or above an unconstrained MLP baseline.
- Identified the task-coupling OOD penalty on CIC-IoT-2023, where the degree of coupling between concept learning and the classification task dictates open-set detectability.

Sneak Path Mitigation in a Crossbar-Based AI Accelerator 2024 – 2025
Southern Illinois University Carbondale

- Investigated the sneak path problem in resistive memristor crossbar arrays used for in-memory neural network inference.
- Designed a 1D1R (diode + memristor) cell in Cadence Virtuoso, reducing read error from 3,430% to under 10%.

Full-Custom CMOS XOR Gate Layout and Verification 2024 – 2025
Southern Illinois University Carbondale

- Designed a 2-input XOR gate from schematic to physical layout in 45 nm GPDK45 using a complementary pass-transistor topology in Cadence Virtuoso.
- Achieved zero DRC and LVS violations across the full RTL-to-GDS flow.

Wireless Localization on Constrained Hardware (B.Tech Thesis) Jul 2023 – Apr 2024
GITAM School of Technology

- Designed a real-time Wi-Fi fingerprint localization framework for IoT nodes, achieving 99% accuracy with an average response time of 3 seconds.
- Addressed self-localization for resource-limited devices in asset tracking and indoor navigation.
- Published at IEEE CONECCT 2024.

Teaching & Academic Service

Graduate Teaching Assistant — Southern Illinois University Carbondale Aug 2025 – Present

- ECE 508: Computer System Security (Fall 2026)
- ECE 513: Digital VLSI Design (Spring 2026) — lab instruction, grading, office hours
- ECE 433: Networking (Fall 2025) — assignment grading and student support

Professional Experience

Afrotel Group Telecommunication Engineering | Network Engineer Jun 2024 – Jul 2025

- Installed and maintained LTE, 5G RAN, microwave, and FTTX fiber optic infrastructure.
- Conducted OTDR, CD/PMD, and power meter tests; performed splice quality validation and point-of-break analysis.
- Troubleshot network faults to prevent downtime; managed MPLS data center connectivity.

EduSkills Foundation | Embedded Systems Intern May 2023 – Jul 2023

- Developed and tested embedded hardware/software solutions integrating sensors, actuators, and communication modules.
- Performed circuit-level troubleshooting on resistive, capacitive, and diode components.

South Eastern Kenya University | ICT Technician Sep 2019 – Feb 2021

- Managed network infrastructure, computer labs, and CCTV systems; provided ICT training for students and staff.

Technical Skills

EDA & Design	Cadence Virtuoso, SPICE simulation, 45 nm GPDK45
HDL	VHDL, Verilog, FPGA/CPLD
Programming	Python, MATLAB, Simulink
Networking	LTE/5G, FTTX, MPLS, RSSI/Wi-Fi fingerprinting
Hardware	Arduino, Raspberry Pi, NodeMCU

Awards & Scholarships

- Study in India Merit-Based Scholarship (2020–2024)